Result:- Let $T: U \to V$ be a linear map, then

(i) If T is one-one and $u_1, u_2, \ldots, u_n$ are LI vectors of U, then $T(u_1), T(u_2), \ldots, T(u_n)$ are LI.

(ii) If $v_1, v_2, \ldots, v_n$ are LI vectors of $R(T)$ and $u_1, u_2, \ldots, u_n$ are vectors of U st $T(u_i) = v_i$, $i = 1, 2, \ldots, n$ then $u_1, u_2, \ldots, u_n$ are LI.

* $\dim(\text{Ker } T) = \text{Nullity}(T)$

$\dim(R(T)) = \text{rank}(T)$

Rank-Nullity Theorem:-

Let $T: U \to V$ be a linear map and U is finite dimensional v.s, then $\dim(U) = \text{rank}(T) + \text{nullity}(T)$

$$\dim(U) = r(T) + n(T)$$

Proof:-

Let $\dim(N(T)) = n(T) = n$

$\dim(U) = p$

Suppose $B = \{u_1, u_2, \ldots, u_n\}$ is a basis of $N(T)$

i.e. $T(u_i) = 0$, $i = 1, 2, \ldots, n$

To get the basis of U, we can extend the set B

$\{u_1, u_2, \ldots, u_n, u_{n+1}, u_{n+2}, \ldots, u_p\}$ is the basis of U.

Let $A = \{T(u_{n+1}), T(u_{n+2}), \ldots, T(u_p)\}$ is the basis of $R(T)$

i.e. $R(T) = [A]$

$$\alpha_{n+1} T(u_{n+1}) + \alpha_{n+2} T(u_{n+2}) + \ldots + \alpha_p T(u_p) = 0$$

$$\Rightarrow T(\alpha_{n+1} u_{n+1} + \alpha_{n+2} u_{n+2} + \ldots + \alpha_p u_p) = 0 \quad [\because T \text{ is linear}]$$

$$\Rightarrow \alpha_{n+1} u_{n+1} + \alpha_{n+2} u_{n+2} + \ldots + \alpha_p u_p \in N(T)$$

$$\Rightarrow \alpha_{n+1} u_{n+1} + \alpha_{n+2} u_{n+2} + \ldots + \alpha_p u_p = \alpha_1 u_1 + \alpha_2 u_2 + \ldots + \alpha_n u_n$$

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